INTERNET PROGRAMMING LAB

Experiment No. 10

Title: Online Fashion E-Commerce Website (StyleHub)

AIM

To design and develop an online fashion store using HTML, CSS and JavaScript with categories for men, women and children, size selection, size guide, flash-sale discounts, wish list, shopping cart, checkout, order tracking with return/exchange requests, and an administrator panel to manage items and view customer details.

ALGORITHM

Step 1. Start and display the Home page with the flash-sale countdown timer, categories (Men, Women, Kids, Accessories) and sub-categories.

Step 2. Update the countdown every second until the sale ends.

Step 3. Search / Filter: read category, sub-category, search text, maximum price and sort order; compute the sale price = price x (100 - discount)/100 and display matching products.

Step 4. Size Selection: the customer selects a size (S, M, L, XL) for each garment; the Size Guide page shows measurements and the return policy.

Step 5. Add to Cart: if no size is selected show a message; otherwise add item with size and quantity.

Step 6. Wish list: logged-in users can add/remove favourite items.

Step 7. Cart: display items, edit quantity, remove items and compute the total using the sale prices.

Step 8. Checkout: validate login, name and address; create the order with status "Processing" and show the confirmation.

Step 9. Orders: display order tracking; for delivered orders allow a return/exchange request.

Step 10. Admin: add new items, remove items, modify price and description, view customers who checked out and update order status.

Step 11. Logout and stop.

PROGRAM

File: p10.html (HTML structure, CSS styling and JavaScript logic in a single file)

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<title>StyleHub - Online Fashion Store</title>
<style>
:root{--c:#c2185b}
*{box-sizing:border-box;margin:0;padding:0;font-family:Segoe UI,Arial,sans-serif}
body{background:#f2f5fa;color:#222;line-height:1.4}
header{background:var(--c);color:#fff;padding:12px 20px;display:flex;justify-content:space-between;align-
items:center;flex-wrap:wrap;gap:8px}
nav button{background:none;border:0;color:#fff;margin-left:4px;padding:7px 11px;cursor:pointer;border-
radius:4px;font-size:14px}
nav button:hover{background:#ffffff33}
main{max-width:1000px;margin:18px auto;padding:0 12px}
.card{background:#fff;border-radius:8px;padding:16px;margin-bottom:16px;box-shadow:0 1px 4px #0002}
.page{display:none}.page.on{display:block}
h3{margin-bottom:10px;color:var(--c)}
input,select,textarea{padding:7px;margin:4px 6px 4px 0;border:1px solid #bbb;border-radius:4px;font-
size:14px}
.btn{background:var(--c);color:#fff;border:0;padding:8px 14px;border-radius:4px;cursor:pointer;font-
size:14px;margin:2px}
.btn.red{background:#b3261e}.btn.gray{background:#666}
table{width:100%;border-collapse:collapse;margin-top:8px;font-size:14px}
th,td{border:1px solid #ddd;padding:7px;text-align:left}th{background:#e9eef6}
.ok{color:#0a7a2f;margin:6px 0}.err{color:#b3261e;margin:6px 0}
.grid{display:grid;grid-template-columns:repeat(auto-fill,minmax(215px,1fr));gap:12px}
.item{border:1px solid #e1e4ea;border-radius:8px;padding:12px;background:#fff}
.pic{height:90px;border-radius:6px;display:flex;align-items:center;justify-content:center;font-
size:44px;background:linear-gradient(135deg,#dfe9f7,#f7e9df);margin-bottom:8px}
.pill{display:inline-block;background:#e5ecf8;border-radius:12px;padding:2px 10px;font-size:12px;margin:2px}
.stat{display:inline-block;min-width:150px;background:#eef3fb;border-radius:8px;padding:10px
14px;margin:4px;text-align:center}
.stat b{display:block;font-size:22px;color:var(--c)}
@media(max-width:600px){nav button{padding:5px 6px;font-size:12px}table{font-size:12px}}

.price{font-weight:bold;color:#c2185b;font-size:17px}
.old{text-decoration:line-through;color:#888;font-size:13px}
.cat{cursor:pointer}.cat.on{background:#c2185b;color:#fff}
.wish{cursor:pointer;font-size:20px;float:right;color:#c2185b}
.flash{background:#fff0f5;border:2px dashed #c2185b;text-align:center;font-size:16px}
.sz{display:inline-block;border:1px solid #777;border-radius:4px;padding:2px
8px;margin:2px;cursor:pointer;font-size:12px}
.sz.on{background:#c2185b;color:#fff;border-color:#c2185b}
</style>
</head>
<body>
<header>
  <h2>&#128087; StyleHub</h2>
  <nav>
    <button onclick="go('home')">Shop</button>
    <button onclick="go('wish')">Wish List</button>
    <button onclick="go('guide')">Size Guide</button>
    <button onclick="go('cart')">Cart (<span id="cc">0</span>)</button>
    <button onclick="go('orders')">Orders</button>
    <button onclick="go('admin')">Admin</button>
    <button id="authBtn" onclick="authClick()">Login</button>
  </nav>
</header>
<main>
  <section id="login" class="page card">
```

```
    <h3>Login / Register</h3>
    <input id="lu" placeholder="Username"><input id="lp" type="password" placeholder="Password">
    <button class="btn" onclick="login()">Login</button><button class="btn gray"
onclick="reg()">Register</button>
    <p id="lmsg"></p><small>Admin demo login: admin / admin123</small>
  </section>

  <section id="home" class="page">
    <div class="card flash">&amp;#9889; FLASH SALE &amp;mdash; extra discounts on marked items! Ends in <b
id="timer">--:--:--</b></div>
    <div class="card">
      <h3>Browse Collections</h3><div id="cats"></div><div id="subs" style="margin-top:6px"></div>
      <input id="q" placeholder="Search (e.g. shirt)" oninput="renderHome()">
      Max price <input type="number" id="maxp" placeholder="Rs." style="width:90px" oninput="renderHome()">
      Sort <select id="sort" onchange="renderHome()"><option value="pop">Popular</option><option
value="lo">Price: low to high</option><option value="hi">Price: high to low</option></select>
    </div>
    <div class="card"><h3 id="ptitle">Products</h3><div id="plist" class="grid"></div></div>
  </section>

  <section id="wish" class="page card"><h3>My Wish List</h3><div id="wlist" class="grid"></div></section>

  <section id="guide" class="page card">
    <h3>Size Guide (cm)</h3>
    <table><tr><th>Size</th><th>Chest</th><th>Waist</th><th>Hip</th></tr>
      <tr><td>S</td><td>86-91</td><td>71-76</td><td>89-94</td></tr><tr><td>M</td><td>96-101</td><td>81-86</
td><td>99-104</td></tr>
      <tr><td>L</td><td>106-111</td><td>91-96</td><td>109-114</td></tr><tr><td>XL</td><td>116-121</
td><td>101-106</td><td>119-124</td></tr></table>
    <p style="margin-top:8px"><b>Return &amp;amp; Exchange policy:</b> returns or size exchanges accepted within
15 days of delivery for unused items with tags.</p>
  </section>

  <section id="cart" class="page card">
    <h3>Shopping Cart</h3><div id="clist"></div>
    <div id="co" style="display:none;margin-top:10px">
      <input id="cname" placeholder="Full name"><input id="cadd" placeholder="Delivery address"
style="width:240px">
      <select id="cpay"><option>Credit/Debit Card</option><option>UPI</option><option>Net
Banking</option><option>Cash on Delivery</option></select>
      <button class="btn" onclick="checkout()">Place Order</button>
    </div>
    <p id="cmsg"></p>
  </section>

  <section id="orders" class="page card"><h3>Order Tracking &amp;amp; History</h3><div
id="olist"></div></section>

  <section id="admin" class="page card">
    <h3>Admin Panel</h3>
    <input id="an" placeholder="Item name"><select id="ac"></select><input id="asub" placeholder="Sub-
category" style="width:110px">
    <input id="ap" type="number" placeholder="Price" style="width:80px"><input id="ad"
placeholder="Description" style="width:190px">
    <button class="btn" onclick="addItem()">Add Item</button><p id="amsg"></p>
    <div id="atable"></div>
    <h3 style="margin-top:14px">Customers who checked out</h3><div id="ctable"></div>
  </section>
</main>

<script>
const $ = id => document.getElementById(id);
const load = (k, d) => { try { return JSON.parse(localStorage.getItem(k)) || d; } catch (e) { return d; } };
const save = (k, v) => { try { localStorage.setItem(k, JSON.stringify(v)); } catch (e) {} };

const CATS = {Men:['Casual wear','Formal attire','Footwear'], Women:['Casual wear','Formal
attire','Seasonal'], Kids:['Casual wear','Seasonal'], Accessories:['Bags','Watches']};
const SIZES = ['S', 'M', 'L', 'XL'];
let users = load('sh_users', [{u:'admin', p:'admin123', role:'admin'}]);
```

```
let items = load('sh_items', [
  {id:1, name:'Classic Denim Shirt',   cat:'Men',   sub:'Casual wear',  price:1499, sale:20, pop:90,
icon:'&#128085;', desc:'100% cotton slim-fit denim shirt.'},
  {id:2, name:'Slim Fit Formal Suit',  cat:'Men',   sub:'Formal attire',price:6999, sale:0,  pop:75,
icon:'&#128084;', desc:'Two-piece suit, wrinkle resistant.'},
  {id:3, name:'Floral Summer Dress',   cat:'Women', sub:'Seasonal',     price:1899, sale:30, pop:95,
icon:'&#128087;', desc:'Light breathable floral midi dress.'},
  {id:4, name:'Office Blazer',         cat:'Women', sub:'Formal attire',price:3499, sale:0,  pop:80,
icon:'&#129509;', desc:'Tailored blazer for work and events.'},
  {id:5, name:'Kids Winter Jacket',    cat:'Kids',  sub:'Seasonal',     price:1299, sale:15, pop:70,
icon:'&#129509;', desc:'Warm padded jacket with hood.'},
  {id:6, name:'Running Sneakers',      cat:'Men',   sub:'Footwear',     price:2999, sale:10, pop:88,
icon:'&#128095;', desc:'Cushioned sole, breathable mesh.'},
  {id:7, name:'Leather Handbag',       cat:'Accessories', sub:'Bags',   price:2499, sale:0,  pop:65,
icon:'&#128092;', desc:'Genuine leather, three compartments.'}
]);
let orders = load('sh_orders', []), wish = load('sh_wish', []), cart = [], sizes = {}, me = null, catSel =
'All', subSel = '';
const sp = i => Math.round(i.price * (100 - i.sale) / 100);

function msg(id, t, bad) { $(id).className = bad ? 'err' : 'ok'; $(id).textContent = t; }
function go(p) {
  if (['orders', 'wish'].includes(p) && !me) { msg('lmsg', 'Please login first.', 1); p = 'login'; }
  if (p === 'admin' && (!me || me.role !== 'admin')) { msg('lmsg', 'Admin login required.', 1); p = 'login';
}
  document.querySelectorAll('.page').forEach(s => s.classList.remove('on')); $(p).classList.add('on');
  if (p === 'home') renderHome(); if (p === 'cart') renderCart(); if (p === 'orders') renderOrders();
  if (p === 'admin') renderAdmin(); if (p === 'wish') $('wlist').innerHTML = items.filter(i =>
wish.includes(i.id)).map(card).join('') || '<p>Wish list is empty.</p>';
}
function reg() {
  const u = $('lu').value.trim(), p = $('lp').value;
  if (!u || p.length < 4) return msg('lmsg', 'Username and 4+ char password required.', 1);
  if (users.some(x => x.u === u)) return msg('lmsg', 'User exists.', 1);
  users.push({u, p, role:'customer'}); save('sh_users', users); msg('lmsg', 'Registered. Please login.');
}
function login() {
  const f = users.find(x => x.u === $('lu').value.trim() && x.p === $('lp').value);
  if (!f) return msg('lmsg', 'Invalid credentials.', 1);
  me = f; $('authBtn').textContent = 'Logout (' + f.u + ')'; go(f.role === 'admin' ? 'admin' : 'home');
}
function authClick() { if (me) { me = null; cart = []; $('cc').textContent = 0; $('authBtn').textContent =
'Login'; } go('login'); }

/* flash-sale countdown */
let end = Date.now() + 3 * 3600 * 1000 + 25 * 60 * 1000;
function tick() { const s = Math.max(0, Math.floor((end - Date.now()) / 1000)), p = n =>
String(n).padStart(2, '0');
  $('timer').textContent = `${p(Math.floor(s / 3600))}:${p(Math.floor(s % 3600 / 60))}:${p(s % 60)}`; }
tick(); setInterval(tick, 1000);

/* ---------- catalogue ---------- */
function renderHome() {
  $('cats').innerHTML = ['All', ...Object.keys(CATS)].map(c => `<button class="btn gray cat ${c === catSel ?
'on' : ''}" onclick="catSel='${c}';subSel='';renderHome()">${c}</button>`).join('');
  $('subs').innerHTML = catSel !== 'All' ? CATS[catSel].map(s => `<span class="pill cat ${s === subSel ?
'on' : ''}" onclick="subSel='${s}';renderHome()">${s}</span>`).join('') : '<small>Sub-categories: ' +
[...new Set(Object.values(CATS).flat())].join(' &bull; ') + '</small>';
  const q = $('q').value.toLowerCase(), mp = +$('maxp').value || Infinity, s = $('sort').value;
  const list = items.filter(i => (catSel === 'All' || i.cat === catSel) && (!subSel || i.sub === subSel) &&
sp(i) <= mp && i.name.toLowerCase().includes(q))
    .sort((a, b) => s === 'lo' ? sp(a) - sp(b) : s === 'hi' ? sp(b) - sp(a) : b.pop - a.pop);
  $('ptitle').textContent = 'Products (' + list.length + ')';
  $('plist').innerHTML = list.map(card).join('') || '<p>No items match.</p>';
}
function card(i) {
  const needSize = i.cat !== 'Accessories';
  return `<div class="item"><span class="wish" onclick="toggleWish(${i.id})">${wish.includes(i.id) ?
'&#9829;' : '&#9825;'}</span><div class="pic">${i.icon}</div>
```

```
    <b>${i.name}</b><br><span class="pill">${i.cat} / ${i.sub}</span><br>
    <span class="price">Rs. ${sp(i)}</span> ${i.sale ? `<span class="old">Rs. ${i.price}</span> <span
class="pill" style="background:#ffd6e5">${i.sale}% OFF</span>` : ''}<br>
    <small>${i.desc}</small><br>
    ${needSize ? SIZES.map(z => `<span class="sz ${sizes[i.id] === z ? 'on' : ''}" onclick="sizes[$
{i.id}]='${z}';renderHome();go2()">${z}</span>`).join('') + '<br>' : ''}
    <button class="btn" onclick="addCart(${i.id})">Add to Cart</button></div>`;
}
function go2() { if ($('wish').classList.contains('on')) go('wish'); }
function toggleWish(id) {
  if (!me) { msg('lmsg', 'Login to use the wish list.', 1); return go('login'); }
  wish = wish.includes(id) ? wish.filter(x => x !== id) : [...wish, id]; save('sh_wish', wish); $
('wish').classList.contains('on') ? go('wish') : renderHome();
}

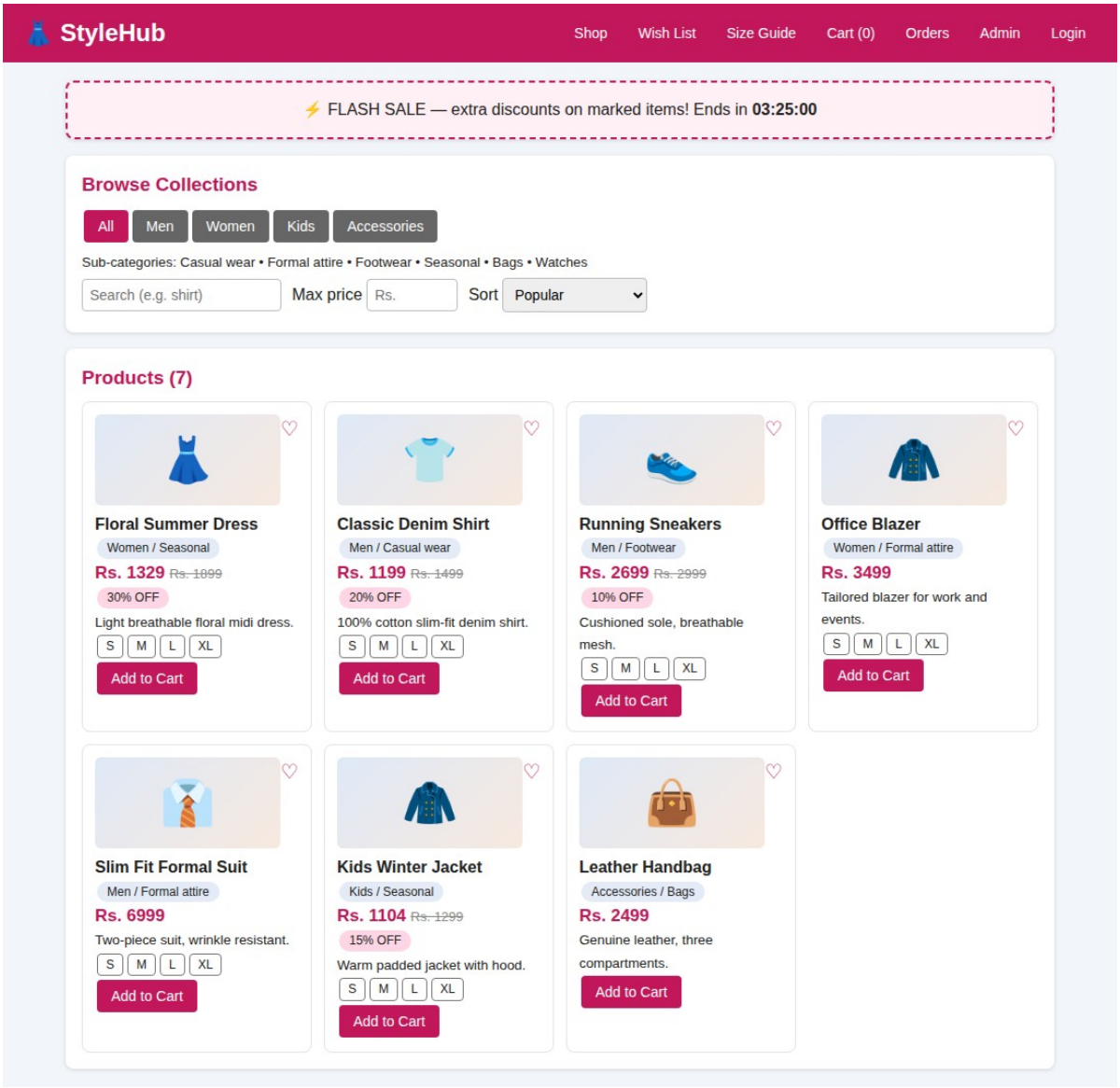
/* ---------- cart ---------- */
function addCart(id) {
  const i = items.find(x => x.id === id), size = i.cat === 'Accessories' ? '-' : sizes[id];
  if (!size) { alert('Please select a size first (see Size Guide).'); return; }
  const l = cart.find(c => c.id === id && c.size === size);
  l ? l.qty++ : cart.push({id, size, qty:1});
  $('cc').textContent = cart.reduce((s, c) => s + c.qty, 0);
}
function renderCart() {
  if (!cart.length) { $('clist').innerHTML = '<p>Your cart is empty.</p>'; $('co').style.display = 'none';
return; }
  let tot = 0;
  $('clist').innerHTML = '<table><tr><th>Item</th><th>Size</th><th>Price</th><th>Qty</th><th>Subtotal</
th><th></th></tr>' + cart.map((c, n) => {
    const i = items.find(x => x.id === c.id); tot += sp(i) * c.qty;
    return `<tr><td>${i.name}</td><td>${c.size}</td><td>${sp(i)}</td><td><input type="number" min="1"
value="${c.qty}" style="width:60px" onchange="cart[${n}].qty=Math.max(1,+this.value||1);renderCart()"></td>
    <td>${sp(i) * c.qty}</td><td><button class="btn red" onclick="cart.splice(${n},1);$
('cc').textContent=cart.length;renderCart()">Remove</button></td></tr>`;
  }).join('') + `<tr><th colspan="4">Total</th><th colspan="2">Rs. ${tot}</th></tr></table>`;
  $('co').style.display = 'block';
}
function checkout() {
  if (!me) { msg('lmsg', 'Please login to check out.', 1); return go('login'); }
  if (!$('cname').value.trim() || !$('cadd').value.trim()) return msg('cmsg', 'Enter name and delivery
address.', 1);
  let tot = 0; const lines = cart.map(c => { const i = items.find(x => x.id === c.id); tot += sp(i) * c.qty;
return {name:i.name, size:c.size, qty:c.qty}; });
  const o = {id:'SH' + Date.now().toString().slice(-6), user:me.u, name:$('cname').value, addr:$
('cadd').value, pay:$('cpay').value, lines, total:tot, status:'Processing'};
  orders.push(o); save('sh_orders', orders); cart = []; $('cc').textContent = 0; renderCart();
  msg('cmsg', `Order ${o.id} placed! Amount Rs. ${o.total}. Track it under Orders.`);
}
function renderOrders() {
  const mine = orders.filter(o => o.user === me.u);
  $('olist').innerHTML = mine.length ? '<table><tr><th>Order</th><th>Items</th><th>Total</th><th>Status</
th><th>Return / Exchange</th></tr>' +
    mine.map(o => `<tr><td>${o.id}</td><td>${o.lines.map(l => `${l.name} (${l.size}) x${l.qty}`).join(',
')}</td><td>Rs. ${o.total}</td><td>${o.status}</td>
    <td>${o.status === 'Delivered' ? `<button class="btn gray" onclick="retReq('${o.id}')">Request
return</button>` : o.status.startsWith('Return') ? 'Requested' : '-'}</td></tr>`).join('') + '</table>' :
'<p>No orders yet.</p>';
}
function retReq(id) { orders.find(o => o.id === id).status = 'Return requested'; save('sh_orders', orders);
renderOrders(); }

/* ---------- admin ---------- */
function renderAdmin() {
  $('ac').innerHTML = Object.keys(CATS).map(c => `<option>${c}</option>`).join('');
  $('atable').innerHTML = '<table><tr><th>Item</th><th>Category</th><th>Price</th><th>Description</
th><th>Actions</th></tr>' + items.map(i =>
    `<tr><td>${i.name}</td><td>${i.cat}/${i.sub}</td><td><input type="number" id="p${i.id}" value="$
{i.price}" style="width:80px"></td>
    <td><input id="d${i.id}" value="${i.desc}" style="width:180px"></td>
```

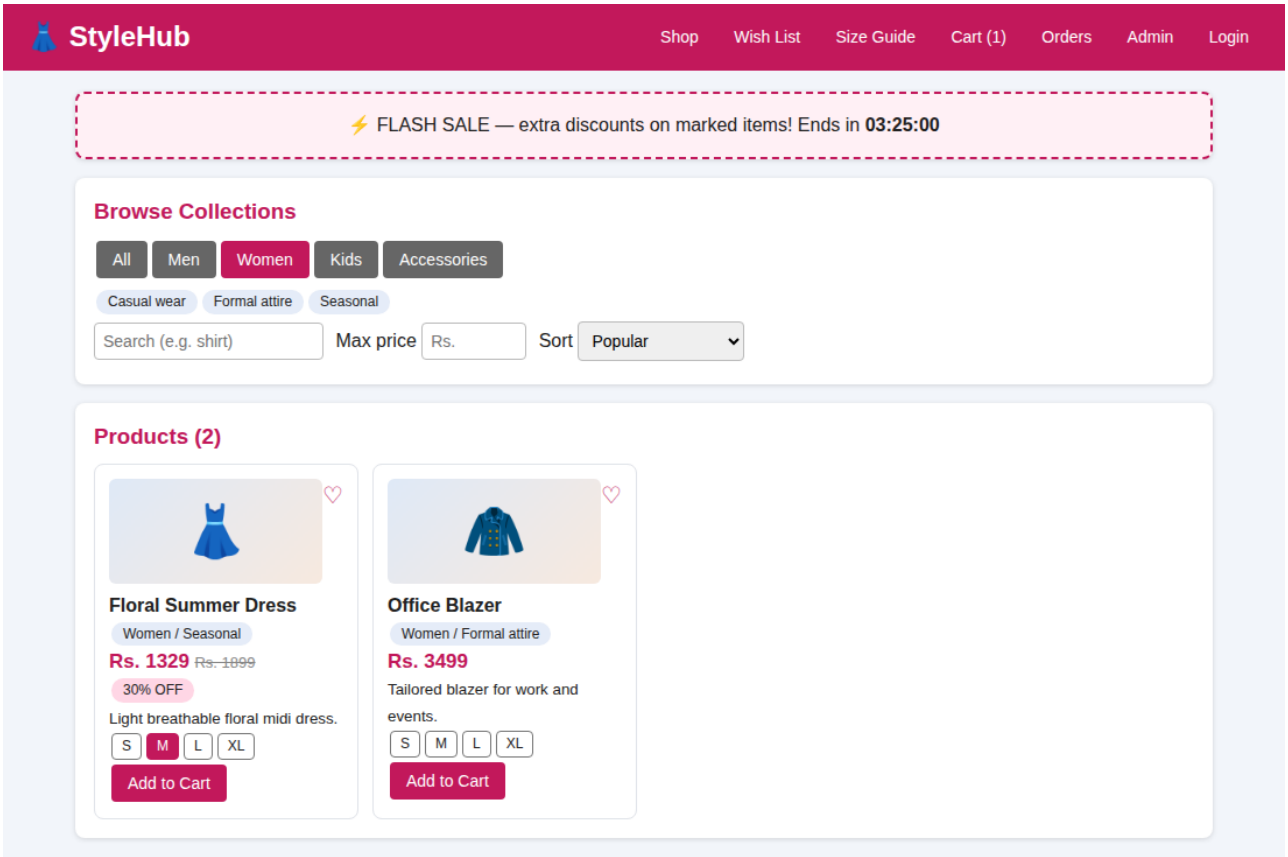
```
    <td><button class="btn" onclick="upd(${i.id})">Update</button><button class="btn red" onclick="del($
{i.id})">Remove</button></td></tr>`).join('') + '</table>';
  $('ctable').innerHTML = orders.length ?
'<table><tr><th>Order</th><th>Customer</th><th>Address</th><th>Total</th><th>Status</th></tr>' +
    orders.map(o => `<tr><td>${o.id}</td><td>${o.name} (${o.user})</td><td>${o.addr}</td><td>${o.total}</td>
    <td><select onchange="setSt('${o.id}',this.value)">${['Processing','Shipped','Delivered'].map(s =>
`<option ${s === o.status ? 'selected' : ''}>${s}</option>`).join('')}</select></td></tr>`).join('') +
'</table>' : '<p>No customer checkouts yet.</p>';
}
function addItem() {
  const it = {id:Date.now(), name:$('an').value.trim(), cat:$('ac').value, sub:$('asub').value.trim() ||
'General', price:+$('ap').value, sale:0, pop:50, icon:'&#128090;', desc:$('ad').value};
  if (!it.name || !it.price) return msg('amsg', 'Name and price required.', 1);
  items.push(it); save('sh_items', items); renderAdmin(); msg('amsg', 'Item added.');
}
function upd(id) { const i = items.find(x => x.id === id); i.price = +$('p' + id).value; i.desc = $('d' +
id).value; save('sh_items', items); msg('amsg', 'Price/description updated.'); }
function del(id) { items = items.filter(i => i.id !== id); save('sh_items', items); renderAdmin();
msg('amsg', 'Item removed.'); }
function setSt(id, s) { orders.find(o => o.id === id).status = s; save('sh_orders', orders); }

go('home');
</script>
</body>
</html>
```

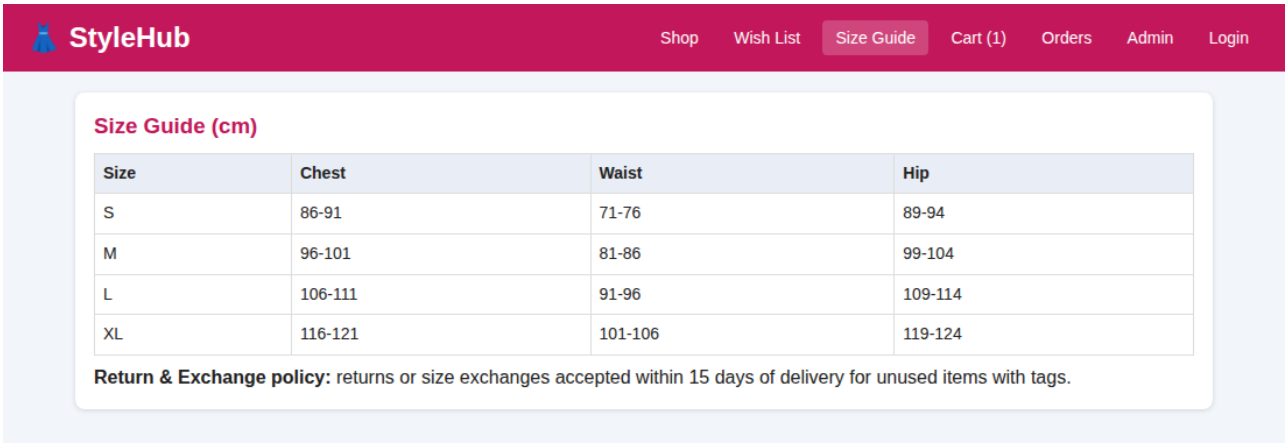
OUTPUT



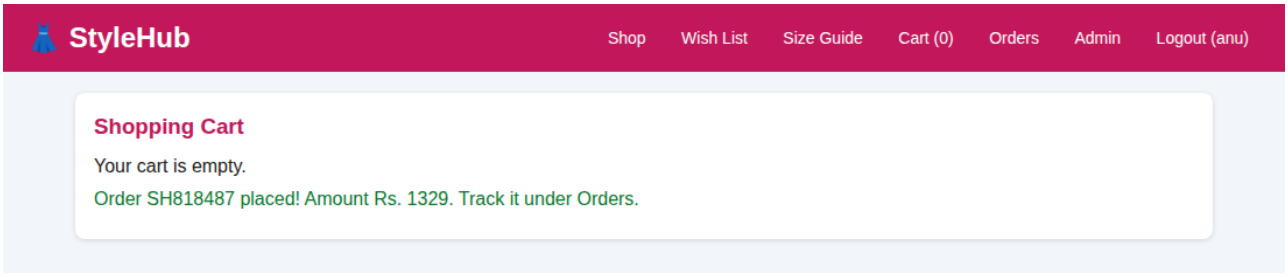
Home page - flash sale timer, categories and product catalogue



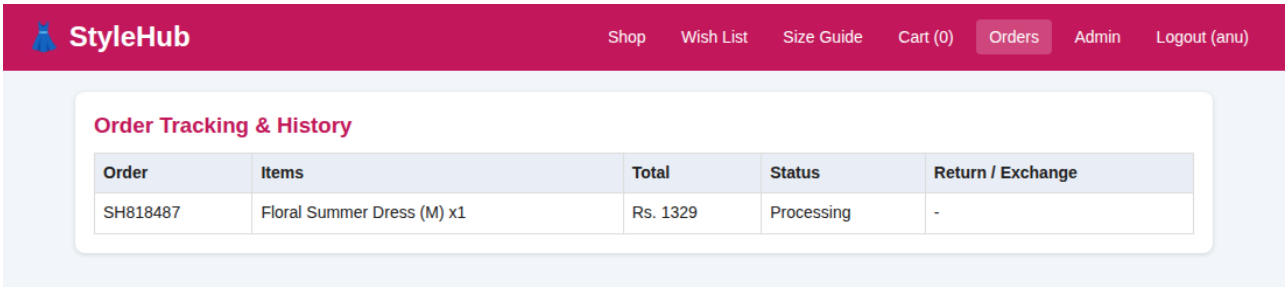
Women category - size selected and item added to cart



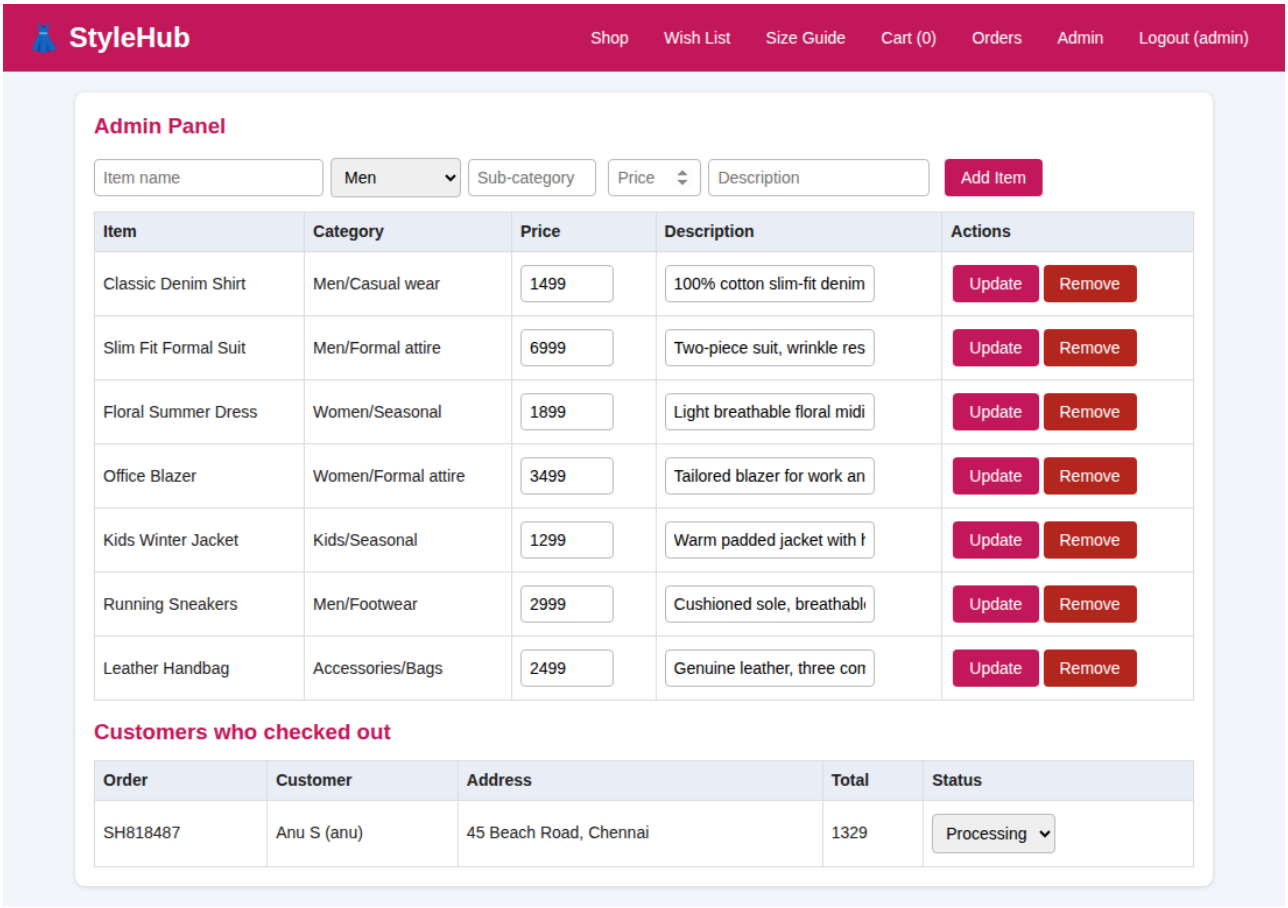
Size guide and return / exchange policy



Order placed successfully



Order tracking page



Admin panel - manage items and customer checkout information

RESULT

Thus the StyleHub fashion e-commerce website was designed and implemented using HTML, CSS and JavaScript. Products were browsed by category, sizes were selected, orders were placed and tracked, and the administrator managed the catalogue. The output was verified successfully.